

$$\phi = 1 + \frac{1}{\phi} = 1 + \frac{1}{1 + \frac{1}{\phi}} = 1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{\phi}}}$$

We can perform this substitution as many times as we like, with no end. As we continue, ϕ gets pushed more and more levels “in” to the corner of a growing fraction. [Listing 5-1](#) shows an expression for ϕ with ϕ seven levels in.

$$\phi = 1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{\phi}}}}}}$$

[Listing 5-1:](#) A continued fraction with seven levels expressing the value of ϕ

If we imagine continuing this process, we can push ϕ infinity levels in. Then what we have left is shown in [Listing 5-2](#).

$$\phi = 1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \dots}}}}}}$$

[Listing 5-2:](#) An infinite continued fraction expressing the value of ϕ